

Program: Diploma in Communication and Computer Networking	
Course Code: 6322D	Course Title: Fundamentals of Networking
Semester: 6	Credits: 1.5
Course Category: Engineering Science	
Periods per week: 3 (L:0 T:3 P:0)	Periods per semester: 45

Course Objectives:

- To understand the basics of computer networking and reference models
- To understand transmission media
- To understand routing protocols
- To understand network layer protocols

Course Prerequisites:

Topic	Course code	Course name	Semester
Fundamentals of computer networks	4322	Computer Networks	4

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Understand the fundamentals of networking and reference models	10	Understanding
CO2	Understand the basic principles of transmission media	11	Understanding
CO3	Understand the design issues of data link layer and error control	10	Applying
CO4	Understand the protocols used in network layer, transport layer and application layer	12	Applying
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2	1		2		
CO2	3	1	2	1	2		
CO3	3	2	1	2			
CO4	3	2	1		2		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Understand the fundamentals of networking and reference models		
M1.01	To understand the uses of computer networks and classification of networks	3	Understanding
M1.02	To understand the network topologies	3	Understanding
M1.03	To study the reference models	4	Understanding
Contents: Introduction of Computer Networks Introduction – Uses of Computer Networks – Classifications of Computer Networks based on scale(PAN, LAN, MAN, WAN and inter networks)- Network Topologies. Reference models – The OSI reference model, The TCP/IP reference model, Comparison of OSI and TCP/IP reference models			
CO2	Understand the basic principles of transmission media		
M2.01	To understand the guided transmission media in computer networks	6	Understanding
M2.02	To study wireless transmission media in computer networks	5	Understanding
	Series Test -I	1	

CO3	Understand the design issues of data link layer and error control		
M3.01	To study error detection mechanisms in data link layer	7	Understanding
M3.02	To study inter connecting devices in data link layer	3	Understanding
Contents: Data link layer: Error Detection and Correction- single bit errors, burst errors- Hamming code, Cyclic Redundancy Code (CRC). Repeaters, Hubs, Bridges and Switches			
CO4	Understand the protocols used in network layer, transport layer and application layer		
M 4.01	To understand routing protocols and IPv4 in network layer	4	Applying
M 4.02	To study the transport layer protocols	4	Understanding
M 4.03	To study the application layer protocols	4	
	Series Test – II	1	
Contents: Network Layer Routing algorithms -Shortest path routing, Flooding, Multicast routing. Internet protocol- IPv4(Addressing and basic concepts only) Transport Layer - Transport Layer Protocols- User Datagram Protocol (UDP). Transmission Control Protocol (TCP) Application Layer - File Transfer Protocol (FTP), Simple Network Management Protocol (SNMP), World Wide Web(WWW) [Basic concepts only]			
Contents: Transmission Media Physical Layer: - Guided Transmission Media– Twisted Pair, Coaxial and Fiber Optics, Wireless Transmission- Radio and Microwave transmission			

Text / Reference

T/R	Book Title/Author
T1	Data Communications and Networking - Behrouz A Forouzan, Tata McGraw-Hill, 5 th edition, ISBN: 9780070634145 for Unit I and III.
T2	Basics of Networking, PHI learning Pvt. Ltd. 2013, ISBN: 978-81-203-2489 for Unit II, IV, V and VI.
T3	Computer Networks is an introductory book written by. Andrew S. Tanenbaum
T4	Data & Computer Communication, Williams Stallings, Prentice Hall of India
R1	Information Technology Today ,S. Jaiswal, Galgotia Publications
R2	Computer Networks , Bhushan Trivedi , Oxford University Press, 2013
R3	Data Communication & Networking, Forouzen ,Tata McGraw Hill
R4	Networks for Computer Scientists and Engineers ,Youlu Zheng & Shakil Akhtar , Oxford University Press, 2012

Online Resources

Sl.No	Website Link
1	http://nptel.iitm.ac.in/courses.php?disciplineId=106 ii.
2	http://www.edrawsoft.com